

ACURON FLEXI

Version 2.1 Revision Date: 09/10/2021 SDS Number: S00042708491 This version replaces all previous versions.

SECTION 1. IDENTIFICATION

Product name : ACURON FLEXI
Design code : A20540D
Product Registration number : 33476
Other means of identification : No data available

Manufacturer or supplier's details

Company name of supplier : Syngenta Canada Inc.
Address : 140 Research Lane, Research Park
Guelph ON N1G 4Z3
Canada

Telephone : 1-87-SYNGENTA (1-877-964-3682)
Telefax : 1-519-823-0504

E-mail address :

Emergency telephone number : 1-800-327-8633 (FAST MED)

Recommended use of the chemical and restrictions on use

Recommended use : Herbicide

SECTION 2. HAZARDS IDENTIFICATION**GHS classification in accordance with the Hazardous Products Regulations**

Skin sensitisation : Category 1
Reproductive toxicity : Category 1B

GHS label elements

Hazard pictograms :



Signal word : Danger

Hazard statements : H317 May cause an allergic skin reaction.
H360 May damage fertility or the unborn child.

Precautionary statements : **Prevention:**
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P261 Avoid breathing mist or vapours.

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P272 Contaminated work clothing should not be allowed out of the workplace.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P302 + P352 IF ON SKIN: Wash with plenty of water.
P308 + P313 IF exposed or concerned: Get medical advice/ attention.
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P362 + P364 Take off contaminated clothing and wash it before reuse.

Storage:

P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	Common Name/Synonym	CAS-No.	Concentration (% w/w)
S-metolachlor	S-metolachlor	87392-12-9	31.242
propane-1,2-diol	propane-1,2-diol	57-55-6	$\geq 1 - < 5$ *
mesotrione (ISO)	mesotrione (ISO)	104206-82-8	3.4712
benoxacor	benoxacor	98730-04-2	$\geq 1 - < 5$ *
nitric acid ammonium salt	nitric acid ammonium salt	6484-52-2	$\geq 1 - < 5$ *
amines, coco alkyl, ethoxylated	amines, coco alkyl, ethoxylated	61791-14-8	$\geq 1 - < 5$ *
calcium dodecylbenzene sulphonate	calcium dodecylbenzene sulphonate	26264-06-2	$\geq 1 - < 5$ *
poly(oxy-1,2-ethanediyl), alpha-tridecyl-omega-hydroxy-, phosphate	poly(oxy-1,2-ethanediyl), alpha-tridecyl-omega-hydroxy-, phosphate	9046-01-9	$\geq 1 - < 5$ *
bicyclopyrone	bicyclopyrone	352010-68-5	0.8676
copper dihydroxide	copper dihydroxide	20427-59-2	$\geq 0.1 - < 1$ *
reaction mass of 5-	reaction mass of	55965-84-9	$\geq 0.0015 - < 0.1$ *

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chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)	5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)		
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* Actual concentration or concentration range is withheld as a trade secret

SECTION 4. FIRST AID MEASURES

General advice	: Have the product container, label or Safety Data Sheet with you when calling the emergency number, a poison control center or physician, or going for treatment.
If inhaled	: Move the victim to fresh air. If breathing is irregular or stopped, administer artificial respiration. Keep patient warm and at rest. Call a physician or poison control centre immediately.
In case of skin contact	: Take off all contaminated clothing immediately. Wash off immediately with plenty of water. If skin irritation persists, call a physician. Wash contaminated clothing before re-use.
In case of eye contact	: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses. Immediate medical attention is required.
If swallowed	: If swallowed, seek medical advice immediately and show this container or label. Do NOT induce vomiting.
Most important symptoms and effects, both acute and delayed	: Nonspecific No symptoms known or expected.
Notes to physician	: There is no specific antidote available. Treat symptomatically.

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media	: Extinguishing media - small fires Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide. Extinguishing media - large fires Alcohol-resistant foam or Water spray
Unsuitable extinguishing media	: Do not use a solid water stream as it may scatter and spread fire.
Specific hazards during fire-fighting	: As the product contains combustible organic components, fire will produce dense black smoke containing hazardous products of combustion (see section 10). Exposure to decomposition products may be a hazard to health.

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Further information : Do not allow run-off from fire fighting to enter drains or water courses.
Cool closed containers exposed to fire with water spray.

Special protective equipment for firefighters : Wear full protective clothing and self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Prevent further leakage or spillage if safe to do so.
Do not flush into surface water or sanitary sewer system.
If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and materials for containment and cleaning up : Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).
Clean contaminated surface thoroughly.
Clean with detergents. Avoid solvents.
Retain and dispose of contaminated wash water.

SECTION 7. HANDLING AND STORAGE

Advice on safe handling : No special protective measures against fire required.
Avoid contact with skin and eyes.
When using do not eat, drink or smoke.
For personal protection see section 8.

Conditions for safe storage : No special storage conditions required.
Keep containers tightly closed in a dry, cool and well-ventilated place.
Keep out of the reach of children.
Keep away from food, drink and animal feedingstuffs.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
S-metolachlor	87392-12-9	TWA	5 mg/m ³	Syngenta
propane-1,2-diol	57-55-6	TWA (Vapour and aerosols)	50 ppm 155 mg/m ³	CA ON OEL
		TWA (aerosol)	10 mg/m ³	CA ON OEL
mesotrione (ISO)	104206-82-8	TWA	5 mg/m ³	Syngenta
benoxacor	98730-04-2	TWA	1 mg/m ³	Syngenta
bicyclopyrone	352010-68-5	TWA	0.7 mg/m ³	Syngenta

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Engineering measures : THE FOLLOWING RECOMMENDATIONS FOR EXPOSURE CONTROLS/PERSONAL PROTECTION ARE INTENDED FOR THE MANUFACTURE, FORMULATION AND PACKAGING OF THE PRODUCT. FOR COMMERCIAL APPLICATIONS AND/OR ON-FARM APPLICATIONS CONSULT THE PRODUCT LABEL.

Containment and/or segregation is the most reliable technical protection measure if exposure cannot be eliminated.

The extent of these protection measures depends on the actual risks in use.

Maintain air concentrations below occupational exposure standards.

Where necessary, seek additional occupational hygiene advice.

Personal protective equipment

Respiratory protection : No personal respiratory protective equipment normally required.
When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hand protection

Remarks : Wear protective gloves. The choice of an appropriate glove does not only depend on its material but also on other quality features and is different from one producer to the other. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time. The break through time depends amongst other things on the material, the thickness and the type of glove and therefore has to be measured for each case. Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Eye protection : No special protective equipment required.

Skin and body protection : Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.
Remove and wash contaminated clothing before re-use.
Wear as appropriate:
Impervious clothing

Protective measures : The use of technical measures should always have priority over the use of personal protective equipment.
When selecting personal protective equipment, seek appropriate professional advice.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

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Appearance	:	suspension
Colour	:	olive green
Odour	:	No data available
Odour Threshold	:	No data available
pH	:	3 - 7 Concentration: 100 % w/v
Melting point/range	:	No data available
Boiling point/boiling range	:	No data available
Flash point	:	Method: Pensky-Martens closed cup does not flash
Evaporation rate	:	No data available
Flammability (solid, gas)	:	No data available
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapour pressure	:	No data available
Relative vapour density	:	No data available
Density	:	1.08 - 1.12 g/cm ³ (20 °C)
Solubility(ies)		
Water solubility	:	No data available
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Auto-ignition temperature	:	425 °C
Decomposition temperature	:	No data available
Viscosity		
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.

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Particle size : No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: None reasonably foreseeable.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: No decomposition if used as directed.
Incompatible materials	: None known.
Hazardous decomposition products	: No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION**Information on likely routes of exposure**

Ingestion
Inhalation
Skin contact
Eye contact

Acute toxicity**Product:**

Acute oral toxicity	: LD50 (Rat, female): > 5,000 mg/kg Remarks: Based on data from similar materials
Acute inhalation toxicity	: LC50 (Rat, male and female): > 2.56 mg/l Exposure time: 4 h Test atmosphere: dust/mist Assessment: The substance or mixture has no acute inhalation toxicity Remarks: Based on data from similar materials
Acute dermal toxicity	: LD50 (Rat, male and female): > 5,000 mg/kg Remarks: Based on data from similar materials

Components:**S-metolachlor:**

Acute oral toxicity	: LD50 (Rat, male and female): 2,672 mg/kg
Acute inhalation toxicity	: LC50 (Rat, male and female): > 2.91 mg/l Exposure time: 4 h Test atmosphere: dust/mist Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	: LD50 (Rabbit): > 2,000 mg/kg Assessment: The substance or mixture has no acute dermal toxicity

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mesotrione (ISO):

- Acute oral toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
- Acute inhalation toxicity : LC50 (Rat, male and female): > 4.75 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Assessment: The substance or mixture has no acute inhalation toxicity
- Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Assessment: The substance or mixture has no acute dermal toxicity

benoxacor:

- Acute oral toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
- Acute inhalation toxicity : LC50 (Rat, male and female): > 2 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Assessment: The substance or mixture has no acute inhalation toxicity
- Acute dermal toxicity : LD50 (Rabbit, male and female): > 2,010 mg/kg
Assessment: The substance or mixture has no acute dermal toxicity

nitric acid ammonium salt:

- Acute oral toxicity : LD50 (Rat): 2,462 mg/kg

amines, coco alkyl, ethoxylated:

- Acute oral toxicity : Assessment: The component/mixture is moderately toxic after single ingestion.

bicyclopyrone:

- Acute oral toxicity : LD50 (Rat, female): > 5,000 mg/kg
- Acute inhalation toxicity : LC50 (Rat, male and female): > 5.21 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Assessment: The substance or mixture has no acute inhalation toxicity
- Acute dermal toxicity : LD50 (Rat, male and female): > 5,000 mg/kg

copper dihydroxide:

- Acute oral toxicity : LD50 (Rat): 451 mg/kg
- Acute inhalation toxicity : LC50 (Rat): 0.50 mg/l

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Exposure time: 4 h
Test atmosphere: dust/mist

reaction mass of 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1):

Acute oral toxicity	: Assessment: The component/mixture is toxic after single ingestion.
Acute inhalation toxicity	: Assessment: The component/mixture is highly toxic after short term inhalation.
Acute dermal toxicity	: Assessment: The component/mixture is highly toxic after single contact with skin.

Skin corrosion/irritation**Product:**

Species	: Rabbit
Result	: No skin irritation
Remarks	: Based on data from similar materials

Components:**S-metolachlor:**

Species	: Rabbit
Result	: No skin irritation

mesotrione (ISO):

Species	: Rabbit
Result	: No skin irritation

benoxacor:

Species	: Rabbit
Result	: No skin irritation

amines, coco alkyl, ethoxylated:

Result	: Corrosive after 3 minutes to 1 hour of exposure
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calcium dodecylbenzene sulphonate:

Result	: Irritating to skin.
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poly(oxy-1,2-ethanediyl), alpha-tridecyl-omega-hydroxy-, phosphate:

Result	: Irritating to skin.
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bicyclopyrone:

Species	: Rabbit
Result	: No skin irritation

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reaction mass of 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1):

Result : Corrosive after 1 to 4 hours of exposure

Serious eye damage/eye irritation**Product:**

Species	: Rabbit
Result	: No eye irritation
Remarks	: Based on data from similar materials

Components:**S-metolachlor:**

Species	: Rabbit
Result	: No eye irritation

mesotrione (ISO):

Species	: Rabbit
Result	: No eye irritation

benoxacor:

Species	: Rabbit
Result	: No eye irritation

nitric acid ammonium salt:

Result : Eye irritation

amines, coco alkyl, ethoxylated:

Result : Irreversible effects on the eye

calcium dodecylbenzene sulphonate:

Result : Risk of serious damage to eyes.

poly(oxy-1,2-ethanediyl), alpha-tridecyl-omega-hydroxy-, phosphate:

Result : Risk of serious damage to eyes.

bicyclopyrone:

Species	: Rabbit
Result	: No eye irritation

copper dihydroxide:

Species	: Rabbit
Result	: Risk of serious damage to eyes.

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Respiratory or skin sensitisation**Product:**

Test Type	: Local lymph node assay (LLNA)
Species	: Mouse
Result	: May cause sensitisation by skin contact.
Remarks	: Based on data from similar materials

Components:**S-metolachlor:**

Species	: Guinea pig
Result	: The product is a skin sensitiser, sub-category 1B.

mesotrione (ISO):

Species	: Guinea pig
Result	: Does not cause skin sensitisation.

benoxacor:

Species	: Guinea pig
Result	: May cause sensitisation by skin contact.

bicyclopyrone:

Test Type	: mouse lymphoma cells
Species	: Mouse
Result	: Did not cause sensitisation on laboratory animals.

reaction mass of 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1):

Result	: The product is a skin sensitiser, sub-category 1A.
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Germ cell mutagenicity**Components:****S-metolachlor:**

Germ cell mutagenicity - Assessment	: Animal testing did not show any mutagenic effects.
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mesotrione (ISO):

Germ cell mutagenicity - Assessment	: Animal testing did not show any mutagenic effects.
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benoxacor:

Germ cell mutagenicity - Assessment	: Animal testing did not show any mutagenic effects.
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bicyclopyrone:

Germ cell mutagenicity - Assessment	: Animal testing did not show any mutagenic effects.
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Assessment

Carcinogenicity**Components:****S-metolachlor:**

Carcinogenicity - Assessment : Animal testing did not show any carcinogenic effects.

mesotrione (ISO):

Carcinogenicity - Assessment : Animal testing did not show any carcinogenic effects.

benoxacor:

Carcinogenicity - Assessment : No evidence of carcinogenicity in animal studies.

bicyclopyrone:

Carcinogenicity - Assessment : This substance has been reported to cause tumours in certain animal species., There is no evidence that these findings are relevant to humans., Weight of evidence does not support classification as a carcinogen

Reproductive toxicity**Components:****S-metolachlor:**

Reproductive toxicity - Assessment : Animal testing did not show any effects on fertility.

mesotrione (ISO):

Reproductive toxicity - Assessment : Animal testing did not show any effects on fertility.

benoxacor:

Reproductive toxicity - Assessment : No toxicity to reproduction

bicyclopyrone:

Reproductive toxicity - Assessment : Animal testing did not show any effects on fertility., These concentrations exceed relevant human dose levels., Clear evidence of adverse effects on development, based on animal experiments.

STOT - repeated exposure**Components:****bicyclopyrone:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

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Repeated dose toxicity**Components:****S-metolachlor:**

Remarks : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

mesotrione (ISO):

Remarks : No adverse effect has been observed in chronic toxicity tests.

benoxacor:

Remarks : No adverse effect has been observed in chronic toxicity tests.

SECTION 12. ECOLOGICAL INFORMATION**Ecotoxicity****Components:****S-metolachlor:**

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 1.23 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Americamysis): 1.4 mg/l
Exposure time: 96 h

Toxicity to algae/aquatic plants : ErC50 (Raphidocelis subcapitata (freshwater green alga)): 0.077 mg/l
Exposure time: 96 h

NOEC (Raphidocelis subcapitata (freshwater green alga)): 0.016 mg/l
End point: Growth rate
Exposure time: 96 h

EC50 (Lemna gibba (gibbous duckweed)): 0.023 mg/l
Exposure time: 14 d

NOEC (Lemna gibba (gibbous duckweed)): 0.0076 mg/l
Exposure time: 14 d

M-Factor (Acute aquatic toxicity) : 10

Toxicity to fish (Chronic toxicity) : NOEC (Pimephales promelas (fathead minnow)): 0.03 mg/l
Exposure time: 35 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Americamysis): 0.13 mg/l
Exposure time: 28 d

M-Factor (Chronic aquatic toxicity) : 10

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toxicity)

mesotrione (ISO):

- Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): > 120 mg/l
Exposure time: 96 h
- LC50 (Cyprinus carpio (Carp)): > 97.1 mg/l
Exposure time: 96 h
- Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 900 mg/l
Exposure time: 48 h
- Toxicity to algae/aquatic plants : ErC50 (Raphidocelis subcapitata (freshwater green alga)): 12 mg/l
Exposure time: 96 h
- NOEC (Raphidocelis subcapitata (freshwater green alga)): 0.75 mg/l
End point: Growth rate
Exposure time: 96 h
- ErC50 (Lemna gibba (gibbous duckweed)): 0.0301 mg/l
Exposure time: 7 d
- EC10 (Lemna gibba (gibbous duckweed)): 0.00187 mg/l
End point: Growth rate
Exposure time: 7 d
- M-Factor (Acute aquatic toxicity) : 10
- Toxicity to fish (Chronic toxicity) : NOEC (Pimephales promelas (fathead minnow)): 12.5 mg/l
Exposure time: 36 d
- Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 180 mg/l
Exposure time: 21 d
- M-Factor (Chronic aquatic toxicity) : 10

Ecotoxicology Assessment

- Acute aquatic toxicity : Very toxic to aquatic life.

benoxacor:

- Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 2.9 mg/l
Exposure time: 96 h
- LC50 (Ictalurus punctatus (channel catfish)): 1.4 mg/l
Exposure time: 96 h
- Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 4.782 mg/l
Exposure time: 48 h
- Toxicity to algae/aquatic : ErC50 (Desmodesmus subspicatus (green algae)): 13.5 mg/l

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plants Exposure time: 72 h

EC10 (Desmodesmus subspicatus (green algae)): 0.22 mg/l
Exposure time: 72 h

Toxicity to fish (Chronic toxicity) : NOEC (Pimephales promelas (fathead minnow)): 0.31 mg/l
Exposure time: 32 d

NOEC (Oncorhynchus mykiss (rainbow trout)): 0.016 mg/l
Exposure time: 21 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 0.354 mg/l
Exposure time: 21 d

amines, coco alkyl, ethoxylated:

Ecotoxicology Assessment

Acute aquatic toxicity : Very toxic to aquatic life.

Chronic aquatic toxicity : Very toxic to aquatic life with long lasting effects.

calcium dodecylbenzene sulphonate:

Ecotoxicology Assessment

Chronic aquatic toxicity : Harmful to aquatic life with long lasting effects.

poly(oxy-1,2-ethanediyl), alpha-tridecyl-omega-hydroxy-, phosphate:

Ecotoxicology Assessment

Acute aquatic toxicity : Toxic to aquatic life.

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

bicyclopyrone:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): > 100 mg/l
Exposure time: 96 h

LC50 (Pimephales promelas (fathead minnow)): > 100 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : ErC50 (Raphidocelis subcapitata (freshwater green alga)): 5.4 mg/l
Exposure time: 96 h

EC10 (Raphidocelis subcapitata (freshwater green alga)): 1.9 mg/l
End point: Growth rate
Exposure time: 96 h

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NOEC (Raphidocelis subcapitata (freshwater green alga)): 1 mg/l
End point: Growth rate
Exposure time: 96 h

ErC50 (Lemna gibba (gibbous duckweed)): 0.055 mg/l
Exposure time: 7 d

NOEC (Lemna gibba (gibbous duckweed)): 0.0032 mg/l
End point: Growth rate
Exposure time: 7 d

M-Factor (Acute aquatic toxicity)	:	10
Toxicity to fish (Chronic toxicity)	:	NOEC (Pimephales promelas (fathead minnow)): 10 mg/l Exposure time: 33 d
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 100 mg/l Exposure time: 21 d
M-Factor (Chronic aquatic toxicity)	:	10
Toxicity to microorganisms	:	EC50 (activated sludge): > 1,000 mg/l Exposure time: 3 h

Ecotoxicology Assessment

Acute aquatic toxicity : Very toxic to aquatic life.

copper dihydroxide:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): > 43.8 mg/l
Exposure time: 96 h

M-Factor (Acute aquatic toxicity) : 10

M-Factor (Chronic aquatic toxicity) : 10

Ecotoxicology Assessment

Acute aquatic toxicity : Very toxic to aquatic life.

Chronic aquatic toxicity : Very toxic to aquatic life with long lasting effects.

reaction mass of 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1):

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 0.22 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia (water flea)): 0.1 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic : ErC50 (Raphidocelis subcapitata (freshwater green alga)):

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plants	0.048 mg/l Exposure time: 72 h NOEC (Raphidocelis subcapitata (freshwater green alga)): 0.0012 mg/l End point: Growth rate Exposure time: 72 h ErC50 (Skeletonema costatum (marine diatom)): 0.0052 mg/l Exposure time: 48 h NOEC (Skeletonema costatum (marine diatom)): 0.00064 mg/l End point: Growth rate Exposure time: 48 h
M-Factor (Acute aquatic toxicity)	: 100
Toxicity to fish (Chronic toxicity)	: NOEC (Oncorhynchus mykiss (rainbow trout)): 0.098 mg/l Exposure time: 28 d
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	: NOEC (Daphnia (water flea)): 0.004 mg/l Exposure time: 21 d
M-Factor (Chronic aquatic toxicity)	: 10

Persistence and degradability**Components:****S-metolachlor:**

Biodegradability	: Result: Not readily biodegradable.
Stability in water	: Degradation half life: 53 - 147 d Remarks: Product is not persistent.

mesotrione (ISO):

Stability in water	: Degradation half life: > 30 d (25 °C) Remarks: Persistent in water.
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benoxacor:

Biodegradability	: Result: Not readily biodegradable.
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bicyclopyrone:

Biodegradability	: Result: Not readily biodegradable.
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reaction mass of 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1):

Biodegradability	: Result: Readily biodegradable.
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Bioaccumulative potential**Components:****S-metolachlor:**

Bioaccumulation : Remarks: Does not bioaccumulate.

Partition coefficient: n-octanol/water : log Pow: 3.05 (25 °C)

mesotrione (ISO):

Bioaccumulation : Remarks: Low bioaccumulation potential.

benoxacor:

Bioaccumulation : Remarks: Does not bioaccumulate.

Partition coefficient: n-octanol/water : log Pow: 2.6 (25 °C)

bicyclopyrone:

Bioaccumulation : Remarks: No data available

Partition coefficient: n-octanol/water : log Pow: -1.9 (25 °C)

Mobility in soil**Components:****S-metolachlor:**

Distribution among environmental compartments : Remarks: Moderately mobile in soils

Stability in soil : Dissipation time: 12 - 46 d
Percentage dissipation: 50 % (DT50)
Remarks: Product is not persistent.

mesotrione (ISO):

Distribution among environmental compartments : Remarks: Highly mobile in soils

Stability in soil : Dissipation time: 6 - 105 d
Percentage dissipation: 50 % (DT50)
Remarks: Product is not persistent.

benoxacor:

Distribution among environmental compartments : Remarks: Moderately mobile in soils

Stability in soil : Dissipation time: 0.9 - 5.3 d
Percentage dissipation: 50 % (DT50)
Remarks: Product is not persistent.

bicyclopyrone:

Distribution among environmental compartments : Remarks: Very highly mobile in soil.

Stability in soil : Remarks: Product is not persistent.

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Other adverse effects**Components:****mesotrione (ISO):**

Results of PBT and vPvB assessment : This substance is not considered to be persistent, bioaccumulating and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulating (vPvB).

benoxacor:

Results of PBT and vPvB assessment : This substance is not considered to be persistent, bioaccumulating and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulating (vPvB).

bicyclopyrone:

Results of PBT and vPvB assessment : This substance is not considered to be persistent, bioaccumulating and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulating (vPvB).

SECTION 13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues	: Refer to the product label for specific disposal/recycling information Do not contaminate ponds, waterways or ditches with chemical or used container. Do not dispose of waste into sewer. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations.
Contaminated packaging	: Refer to the product label for specific disposal/recycling information Empty remaining contents. Triple rinse containers. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers.

SECTION 14. TRANSPORT INFORMATION**International Regulations****UNRTDG**

UN number	: UN 3082
Proper shipping name	: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (S-METOLACHLOR AND MESOTRIONE)
Class	: 9
Packing group	: III

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Labels : 9

IATA-DGR

UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
(S-METOLACHLOR AND MESOTRIONE)
Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964
Environmentally hazardous : yes

IMDG-Code

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
N.O.S.
(S-METOLACHLOR AND MESOTRIONE)
Class : 9
Packing group : III
Labels : 9
EmS Code : F-A, S-F
Marine pollutant : yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

National Regulations**TDG**

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
N.O.S.
(S-METOLACHLOR AND MESOTRIONE)
Class : 9
Packing group : III
Labels : 9
ERG Code : 171
Marine pollutant : yes(S-METOLACHLOR, MESOTRIONE)
Remarks : Class 9 Exemption from Part 3, Documentation, and Part 4,
Dangerous Goods Safety Marks, if transported solely on land
by road vehicle or railway vehicle.
1.45.1. SOR/2008-34

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

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SECTION 15. REGULATORY INFORMATION

Warning, contains the allergens 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-4-isothiazolin-3-one

Warning, contains the allergen 2-bromo-2-nitropropane-1,3-diol

Read the label, authorised under the Pest Control Products Act, prior to using or handling the pest control product

This chemical is a pest control product registered by Health Canada Pest Management Regulatory Agency and is subject to certain labelling requirements under the Pest Control Products Act. These requirements differ from the classification criteria and hazard information required for GHS-consistent safety data sheets. The following is the hazard information required on the pest control product label: There are Canada-specific environmental requirements for handling, use, and disposal of this pest control product that are indicated on the label.

Caution

Eye irritant

Potential skin sensitiser

- | | | |
|-------------------------------|---|---|
| Canadian PBT Chemicals | : | This product contains the following components on the DSL that are classified as Persistent, Bioaccumulative and/or Toxic (PBT) under CEPA:
octamethylcyclotetrasiloxane [D4]Cyclopentasiloxane, 2,2,4,4,6,6,8,8,10,10-decamethyl- |
| NPRI Components | : | copper dihydroxide
Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified
propan-2-ol
toluene
naphthalene
hydrogen chloride
ethylbenzene
xylene |

The components of this product are reported in the following inventories:

- | | | |
|-----|---|---|
| DSL | : | This product contains the following components that are not on the Canadian DSL nor NDSL.
bicyclopyrone

S-metolachlor

mesotrione (ISO)

Siloxanes and Silicones, di-Me, polymers with silica-1,1,1-trimethyl-N-(trimethylsilyl)silanamine hydrolysis products and trimethylsilyl silicate

benoxacor |
|-----|---|---|

Canadian lists

No substances are subject to a Significant New Activity Notification.

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SECTION 16. OTHER INFORMATION**Full text of other abbreviations**

CA ON OEL : Ontario Table of Occupational Exposure Limits made under the Occupational Health and Safety Act.
CA ON OEL / TWA : Time-Weighted Average Limit (TWA)

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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Date format : mm/dd/yyyy

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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